

OSA: Echocardiography Video Segmentation via Orthogonalized State Update and Anatomical Prior-aware Feature Enhancement

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Problem Setting



Medical

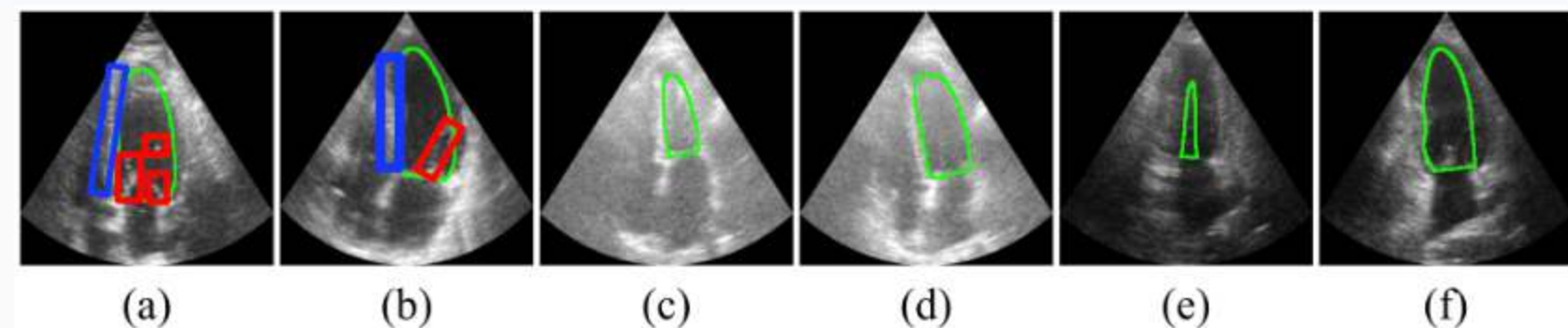
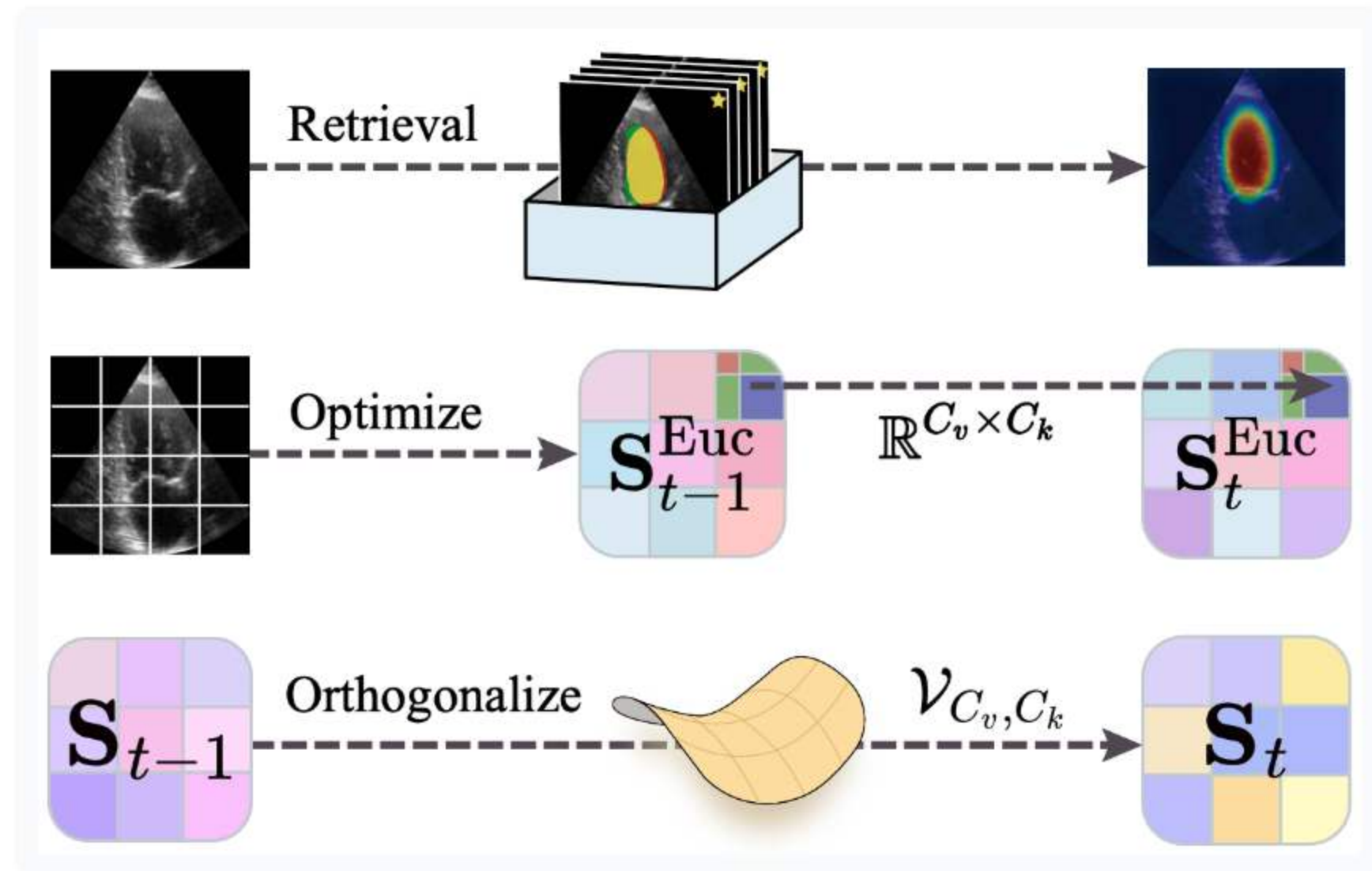
(a) Speckle Noise, (b) Blurred Contours, and (c-f) Pronounced Variations in the Target's Morphology Across the Cardiac Cycle

Video

(a) Extended Temporal Contexts, (b) Efficiency–Accuracy Trade-off in Recall, (c) Computational Burden

Task

Clinical Simulation Setting — Absence of Ground Truth at Inference;
Training via Boundary-Frame Prediction and Loss Computation



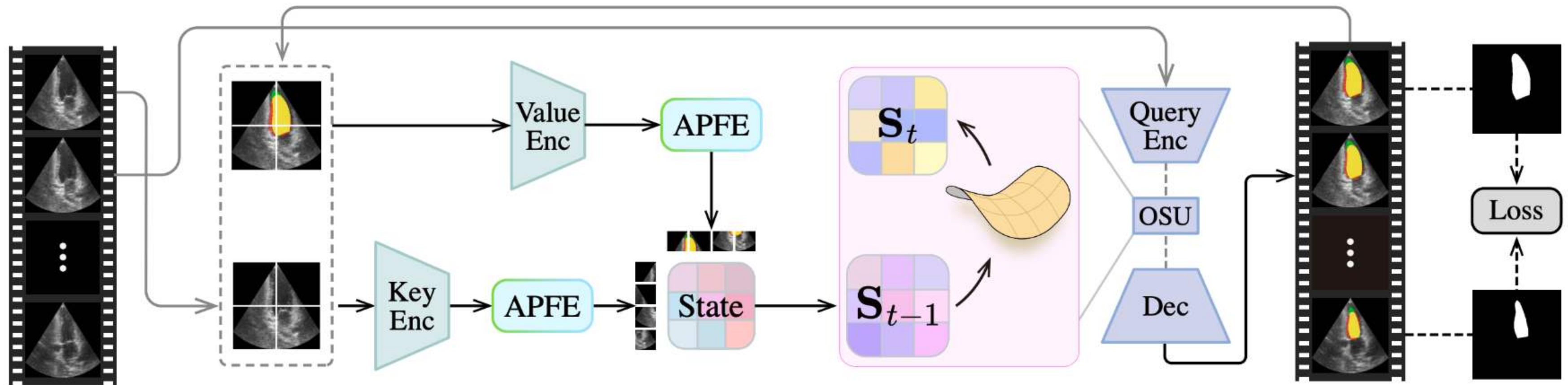
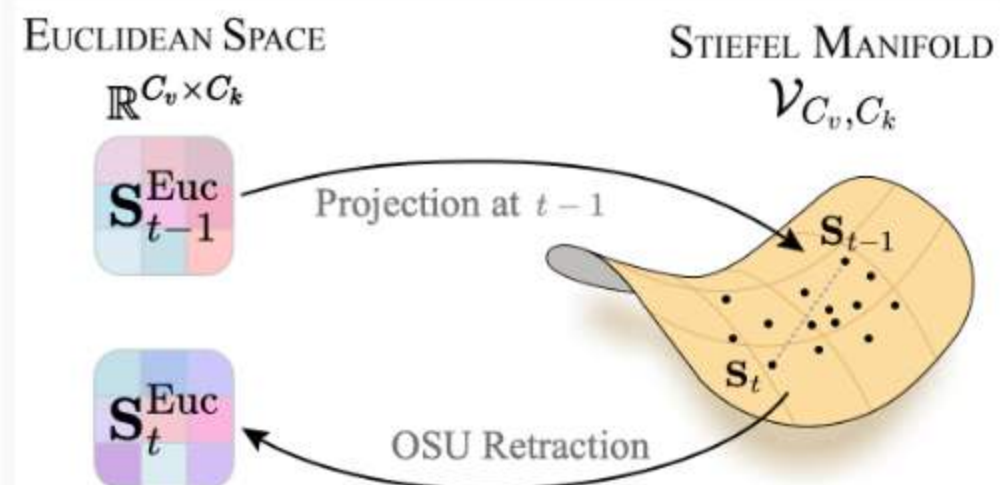


Fig.1: Overall pipeline of OSA — Orthogonalized State Update + Anatomical Prior-aware Feature Enhancement

Method — Orthogonalized State Update

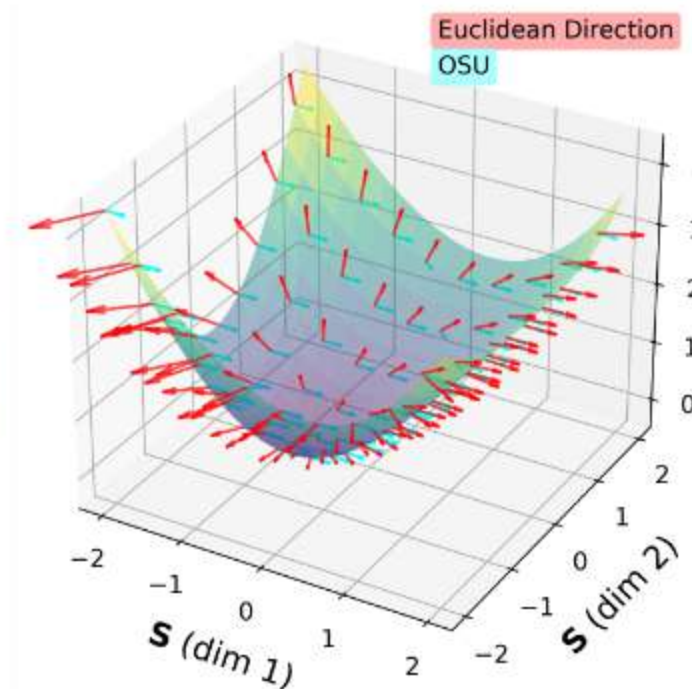
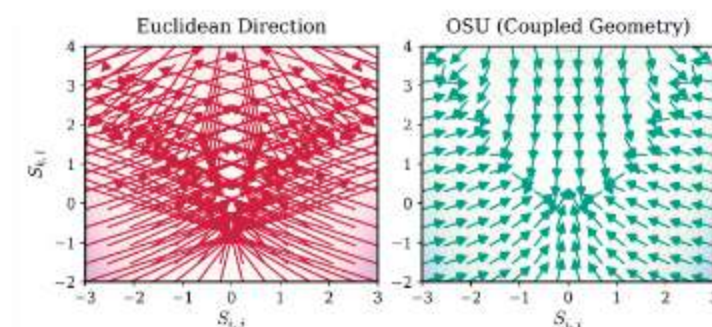


$$\mathbf{S}_t^{\text{Euc}} = \mathbf{S}_{t-1} (\alpha_t (\mathbf{I}_{C_k} - \beta_t \mathbf{k}_t \mathbf{k}_t^\top) + \beta_t \mathbf{v}_t \mathbf{k}_t^\top).$$

$$\mathcal{V}_{C_v, C_k} = \{\mathbf{S} \in \mathbb{R}^{C_v \times C_k} : \mathbf{S}^\top \mathbf{S} = \mathbf{I}_{C_k}\}.$$

$$\mathbf{S}_t = \text{Proj}_{\mathcal{V}}(\mathbf{S}_t^{\text{Euc}}) = \arg \min_{\mathbf{S} \in \mathcal{V}_{C_v, C_k}} \frac{1}{2} \|\mathbf{S} - \mathbf{S}_t^{\text{Euc}}\|_F^2.$$

$$\mathbf{X}^{(0)} = \frac{\mathbf{S}_t^{\text{Euc}}}{\|\mathbf{S}_t^{\text{Euc}}\|_F + \epsilon}.$$



Method — Anatomical Prior-aware Feature Enhancement

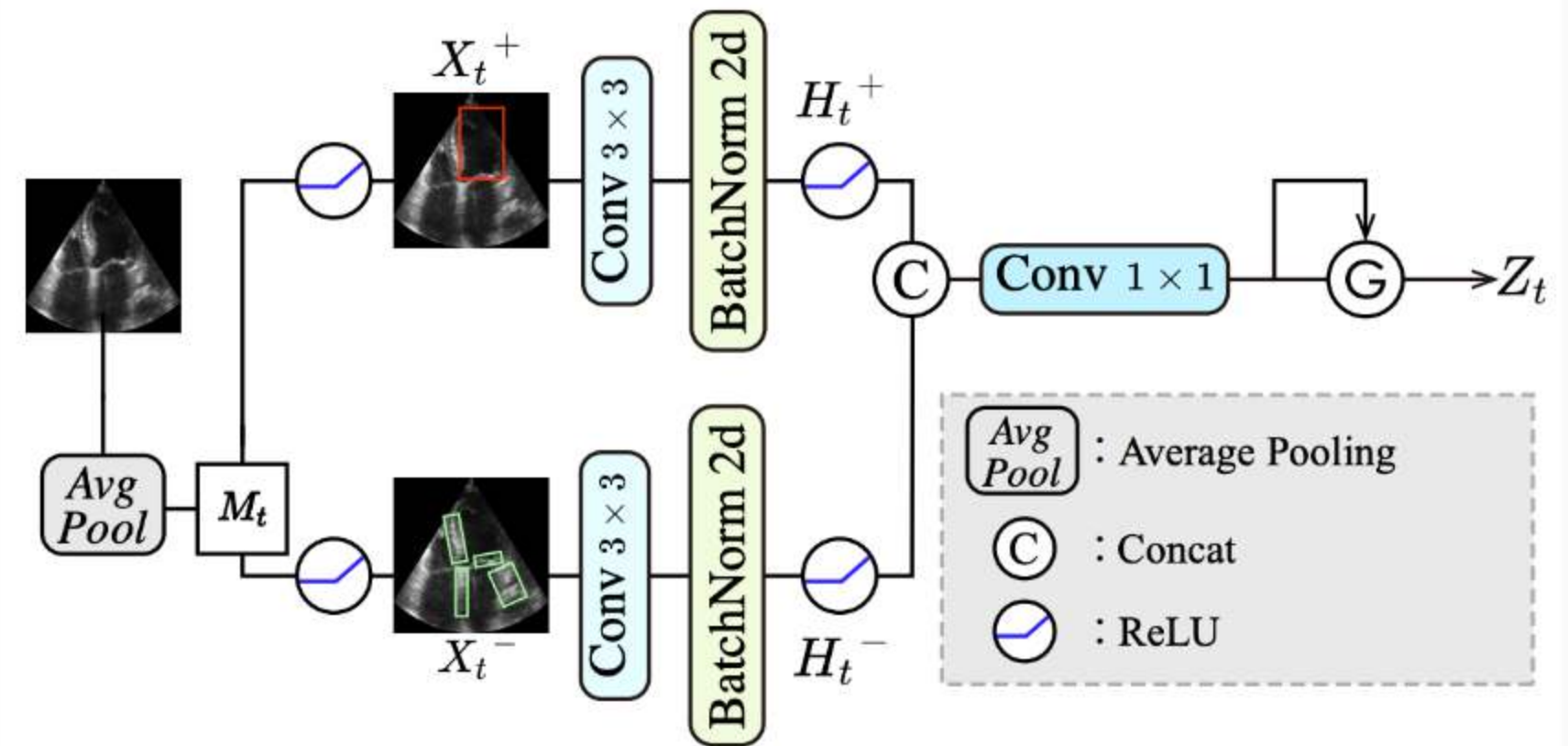
$$\mathbf{M}_t = \text{AvgPool}_{K \times K}(\mathbf{X}_t)$$

$$\mathbf{X}_t^+ = \text{ReLU}(\mathbf{X}_t - \mathbf{M}_t), \quad \mathbf{X}_t^- = \text{ReLU}(\mathbf{M}_t - \mathbf{X}_t)$$

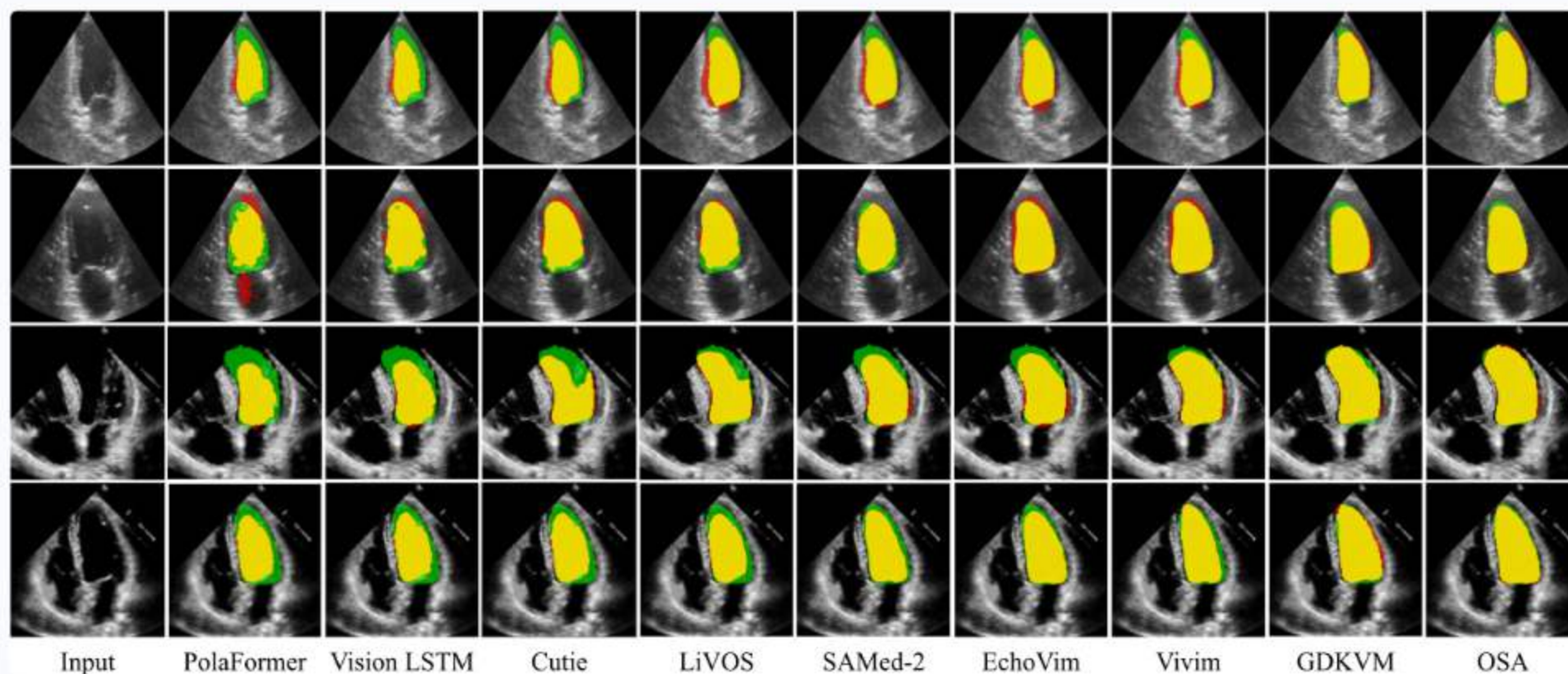
$$\mathbf{H}_t^+ = \phi^+(\mathbf{X}_t^+), \quad \mathbf{H}_t^- = \phi^-(\mathbf{X}_t^-)$$

$$\lambda_t = \sigma(\mathbf{W}_g [\mathbf{H}_t^+; \mathbf{H}_t^-]),$$

$$\mathbf{Z}_t = \lambda_t \odot \mathbf{H}_t^+ + (1 - \lambda_t) \odot \mathbf{H}_t^-$$

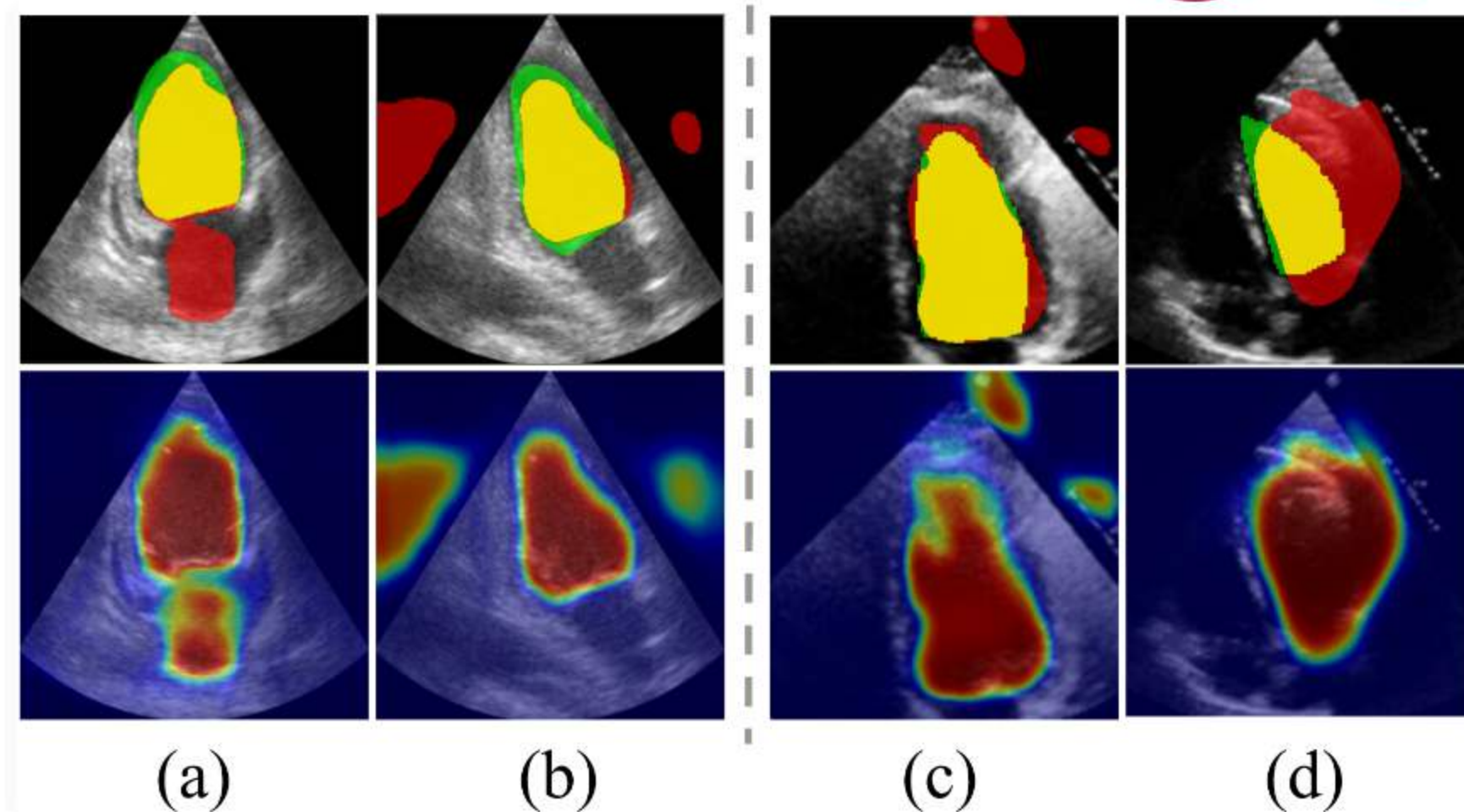
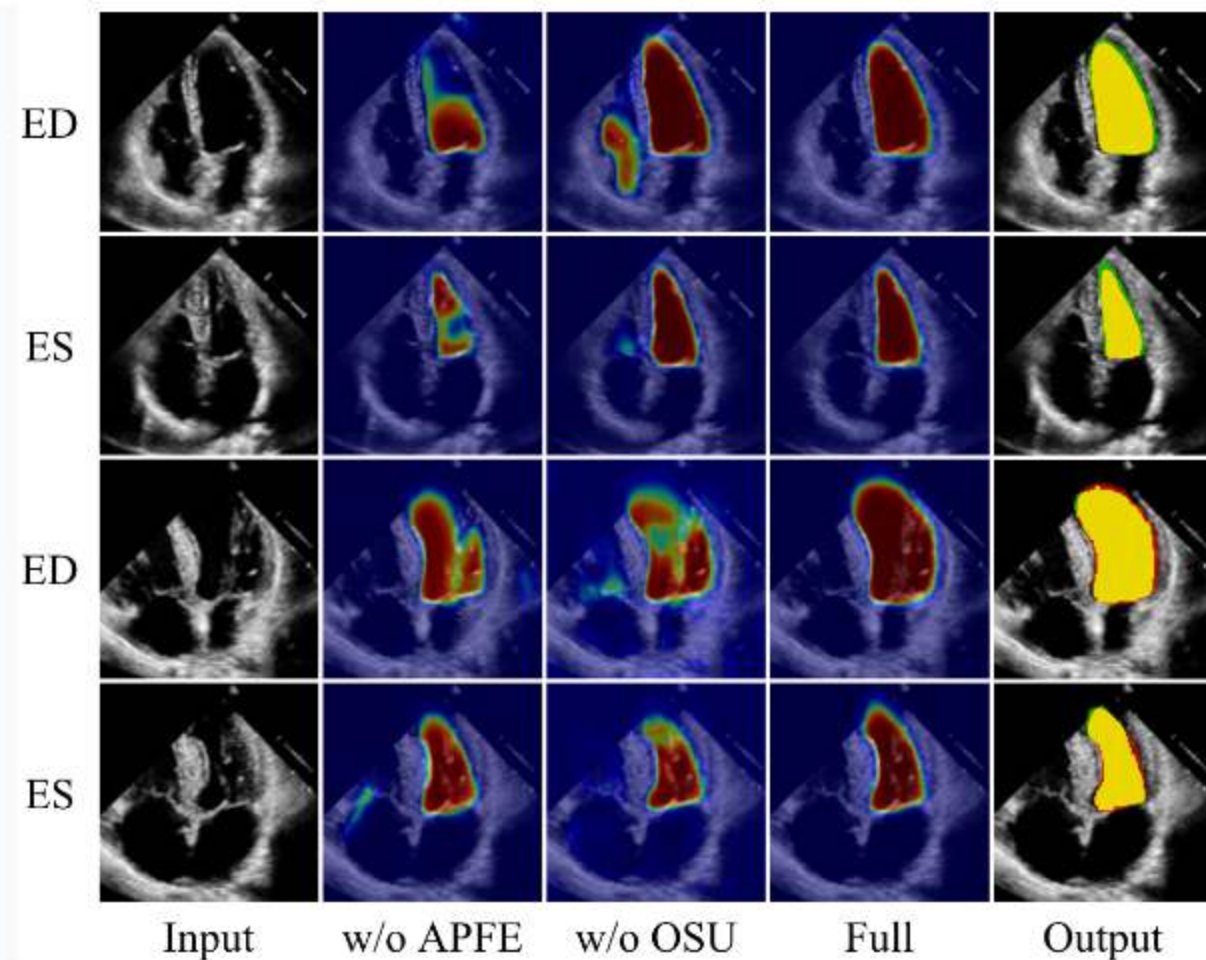


Experiments



- 01** **OSA** achieves state-of-the-art Dice on CAMUS & EchoNet-Dynamic.
- 02** **OSU** stabilizes hidden-state trajectories on Stiefel manifold.
- 03** **APFE** enhances feature discrimination via anatomical priors.
- 04** Lightweight design: only **+3.2%** params vs. baseline.

Experiments



Method	mDice $\uparrow$	mHD95 $\downarrow$	Mem $\downarrow$	Time $\downarrow$
Baseline	92.94	3.56	7.5	4.5
w/o APFE	93.61	3.29	7.6	3.0
w/o OSU	94.12	3.21	7.5	4.7
Full	94.82	3.25	7.6	3.0

- **Generality:** Extend to broader ultrasound datasets.
- **Harder video tasks:** Tackle longer sequences, complex rhythms, and difficult cases.
- **Hardware-aware:** Optimize the matrix state for parallel acceleration.

Thanks!

Paper: [arXiv:2603.26188](#)

Code: github.com/wangrui2025/osa